

Sample Paper 3 — Answer Key

Total: 40 marks (30×1 mark + 5×2 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Spoon** — Pen, Pencil, Eraser are stationery items; a Spoon is not.
2. **B) Sky** — A fish lives in water; a bird's natural space is the sky.
3. **A) HQQV** — The rule is "+2 letters": $F \rightarrow H$, $O \rightarrow Q$, $O \rightarrow Q$, $T \rightarrow V$.
4. **C) 66** — The gaps decrease by 1 each time ($-10, -9, -8, -7$): 100, 90, 81, 73, 66.
5. **D) Uma** — Order: $\text{Uma} > \text{Sana} > \text{Puja} > \text{Rita}$, with Tina the shortest overall.
6. **A) North** — Turning 180° always faces you the opposite way: South $\rightarrow$ North.
7. **B) Thursday** — Sunday (1st), Monday (2nd), Tuesday (3rd), Wednesday (4th), Thursday (5th).
8. **C) 15** — 4, 9, 16 are perfect squares ($2^2, 3^2, 4^2$); 15 is not.
9. **C) 13** — Fibonacci pattern: each term is the sum of the two before it ($5+8=13$).
10. **A) UCPF** — The rule is "+2 letters": $S \rightarrow U$, $A \rightarrow C$, $N \rightarrow P$, $D \rightarrow F$.

Section 2 — Mathematical Reasoning

11. **A) 10112** — $7245 + 2867 = 10112$.
12. **A) 3255** — $6003 - 2748 = 3255$.
13. **A) 1945** — $389 \times 5 = 1945$.
14. **A) 828** — $5796 \div 7 = 828$ exactly.
15. **C) 24** — $24 \div 4 = 6$ and $24 \div 6 = 4$, so it's a multiple of both. (16 isn't a multiple of 6; 18 isn't a multiple of 4; 30 isn't a multiple of 4.)
16. **B) $\frac{2}{3}$** — The greatest common factor of 10 and 15 is 5; $10 \div 5 = 2$, $15 \div 5 = 3$.
17. **B) 0.75** — $\frac{3}{4} = 0.75$.
18. **A) 6 km 200 m** — $6200 \text{ m} = 6000 \text{ m} + 200 \text{ m} = 6 \text{ km } 200 \text{ m}$.
19. **C) 80 sq cm** — Area = length $\times$ breadth = $10 \times 8 = 80 \text{ sq cm}$.
20. **D) 12** — A cube has 12 edges.

Section 3 — Everyday Mathematics

21. **A) 2121** — $1245 + 876 = 2121$.
22. **B) 810** — $18 \times 45 = 810$.

23. **B) 125 cm** — $750 \div 6 = 125$ cm.
24. **B) 10:25 a.m.** — $5:50 \text{ a.m.} + 4 \text{ h} = 9:50 \text{ a.m.}; + 35 \text{ more minutes} = 10:25 \text{ a.m.}$
25. **C) 15** — $9 \text{ L} = 9000 \text{ mL}; 9000 \div 600 = 15$ mugs.
26. **C) ₹95** — $(5 \times ₹12) + ₹35 = ₹60 + ₹35 = ₹95$.
27. **B) 60 L** — $1/5 \times 300 = 60$ L.
28. **C) 1 kg 400 g** — $7300 \text{ g} - 5900 \text{ g} = 1400 \text{ g} = 1 \text{ kg } 400 \text{ g}$.
29. **C) 8** — $96 \div 12 = 8$ trays.
30. **B) ₹250** — $₹2650 - ₹2400 = ₹250$.

Section 4 — Achievers Section (2 marks each)

31. **C) 115** — Number = $(12 \times 9) + 7 = 108 + 7 = 115$.
32. **C) 900** — Let larger = L, smaller = S. $L + S = 1500$, $L - S = 300$. Adding: $2L = 1800 \rightarrow L = 900$.
33. **C) ₹1,500** — Groceries = $1/3 \times 3000 = ₹1000$; remaining = ₹2000; gift = $1/4 \times 2000 = ₹500$; left = $2000 - 500 = ₹1500$.
34. **C) 36** — Erasers = 10; pencils = $10 + 8 = 18$; pens = $2 \times 18 = 36$.
35. **D) 21** — Each term is the sum of the previous two: $8 + 13 = 21$.
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Scoring guide (out of 40): 32+ is Level-2 cutoff range — strong. 24–31 is solid Level-1 performance — review the missed sections. Below 24 — revisit the relevant topics before attempting Sample Paper 4.
